# Linear Algebra and Its Applications

## Section 2.6: The Leontief Input–Output Model

Detailed worked solutions

Mathjock

**Text and numbering.** David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

**Source scope.** The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

**How to read the solutions.** Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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## 2.6 The Leontief input–output model

The production vector $x$ must supply both intermediate consumption $Cx$ and final demand $d$. Thus

$$x = Cx + d, \qquad (I - C)x = d.$$

Column $j$ of $C$ lists the amounts each sector supplies as inputs per unit produced in sector $j$. These columns need not sum to one; the remaining part includes value added. This is a different model from the closed exchange economy of Section 1.6.

The vector $d$ is the external demand, not the total required production. When $I - C$ is invertible, $x = (I - C)^{-1}d$. For a single right side, solving $(I - C)x = d$ by elimination is often more efficient than computing an inverse. Respect the instructions against forming inverses in Exercises 2 and 3 below.

### Practice problem 2.6.P1: Write each unit-consumption vector as a column

The Goods column is $(.2, .5)^T$ and the Services column is $(.4, .3)^T$. Hence

$$x = \begin{bmatrix} .2 & .4 \\ .5 & .3 \end{bmatrix} x + \begin{bmatrix} 20 \\ 30 \end{bmatrix}, \quad \text{or} \quad \begin{bmatrix} .8 & -.4 \\ -.5 & .7 \end{bmatrix} x = \begin{bmatrix} 20 \\ 30 \end{bmatrix}.$$

The first row accounts for all Goods required, and the second accounts for all Services required. This sets up the requested model.

### Exercise 2.6.1: Construct the three-sector consumption matrix

Use Manufacturing, Agriculture, Services as the order of rows and columns. The input columns described in the exercise give

$$C = \begin{bmatrix} .10 & .60 & .60 \\ .30 & .20 & 0 \\ .30 & .10 & .10 \end{bmatrix}.$$

For example producing one agricultural unit requires .60 unit from Manufacturing, .20 from Agriculture, and .10 from Services. If Agriculture produces 100 units, its intermediate-demand vector is 100 times that column:

$$(60, 20, 10)^T.$$

These are the inputs needed for that production plan, not a final-demand vector or a complete equilibrium production plan for all sectors.

### Exercise 2.6.2: Meet agricultural final demand without computing an inverse

The final-demand vector is $(0, 20, 0)^T$. Multiplying the equations $(I - C)x = d$ by 10 gives

$$\left[\begin{array}{rrr|r} 9 & -6 & -6 & 0 \\ -3 & 8 & 0 & 200 \\ -3 & -1 & 9 & 0 \end{array}\right].$$

Swap rows 1 and 2. Then use $R_2 \leftarrow R_2 + 3R_1$, $R_3 \leftarrow R_3 - R_1$, and $R_3 \leftarrow 2R_3 + R_2$. The result is

$$\left[\begin{array}{ccc|c} -3 & 8 & 0 & 200 \\ 0 & 18 & -6 & 600 \\ 0 & 0 & 12 & 200 \end{array}\right].$$

Back substitution gives $x_S = 50/3$, $x_A = (600+6(50/3))/18 = 350/9$, and $x_M = (8(350/9)-200)/3 = 1000/27$.

**Conclusion.** $x = (1000/27, 350/9, 50/3)^T$. Substitution into the original model gives final demand $(0, 20, 0)^T$ exactly.

### Exercise 2.6.3: Meet manufacturing final demand by the same elimination

Now $d = (20, 0, 0)^T$, so the initial augmented right side after multiplying by 10 is $(200, 0, 0)^T$. Use the same row swap and replacements listed in Exercise 2.6.2. They produce

$$\left[\begin{array}{ccc|c} -3 & 8 & 0 & 0 \\ 0 & 18 & -6 & 200 \\ 0 & 0 & 12 & 200 \end{array}\right].$$

Thus $x_S = 50/3$, $x_A = (200 + 100)/18 = 50/3$, and $x_M = 8(50/3)/3 = 400/9$.

**Conclusion.** $x = (400/9, 50/3, 50/3)^T$. No inverse matrix was used.

### Exercise 2.6.4: Add the production responses to two demand vectors

The new demand $(20, 20, 0)^T$ is the sum of the demands in the two previous exercises. If $(I-C)x^{(a)} = d^{(a)}$ and $(I-C)x^{(b)} = d^{(b)}$, additivity gives $(I-C)(x^{(a)}+x^{(b)}) = d^{(a)}+d^{(b)}$. Add the actual production vectors:

$$x = \begin{bmatrix} 1000/27 \\ 350/9 \\ 50/3 \end{bmatrix} + \begin{bmatrix} 400/9 \\ 50/3 \\ 50/3 \end{bmatrix} = \begin{bmatrix} 2200/27 \\ 500/9 \\ 100/3 \end{bmatrix}.$$

The elimination pivots in Exercises 2 and 3 are nonzero, so the coefficient matrix is invertible and this solution is unique.

### Exercise 2.6.5: Use the inverse requested in the problem

Here

$$I - C = \begin{bmatrix} 1 & -1/2 \\ -3/5 & 4/5 \end{bmatrix}, \qquad \det(I-C) = \frac{4}{5} - \frac{3}{10} = \frac{1}{2}.$$

The $2 \times 2$ inverse formula therefore gives

$$(I-C)^{-1} = \begin{bmatrix} 8/5 & 1 \\ 6/5 & 2 \end{bmatrix}.$$

Multiplication by $(50, 30)^T$ yields

$$x = (80 + 30, 60 + 60)^T = (110, 120)^T.$$

Check intermediate demand: $Cx = (60, 90)^T$, so $Cx + d = (60 + 50, 90 + 30)^T = x$.

### Exercise 2.6.6: An inverse with a different determinant

Now

$$I - C = \begin{bmatrix} 4/5 & -1/2 \\ -3/5 & 9/10 \end{bmatrix}, \qquad \det(I - C) = \frac{36}{50} - \frac{3}{10} = \frac{21}{50}.$$

Consequently

$$(I - C)^{-1} = \frac{50}{21} \begin{bmatrix} 9/10 & 1/2 \\ 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 15/7 & 25/21 \\ 10/7 & 40/21 \end{bmatrix}.$$

With $d = (16, 12)^T$, the two coordinates are $240/7 + 100/7 = 340/7$ and $160/7 + 160/7 = 320/7$.

**Conclusion.** $x = (340/7, 320/7)^T$. This vector satisfies $(I - C)x = (16, 12)^T$.

### Exercise 2.6.7: The effect of increasing final demand by one unit

Use the inverse from Exercise 2.6.5.

(a) For demand $(1, 0)^T$, the production vector is the first column of the inverse: $(8/5, 6/5)^T$.

(b) For demand $(51, 30)^T$, multiplication gives $x = ((8/5)51 + 30, (6/5)51 + 60)^T = (558/5, 606/5)^T$.

(c) The new demand is $(50, 30)^T + (1, 0)^T$. By distributivity of matrix multiplication, the new production is $(110, 120)^T + (8/5, 6/5)^T = (558/5, 606/5)^T$. Thus the first inverse column records the whole economy's extra production needed for the one-unit increase in sector 1's final demand.

### Exercise 2.6.8: Prove superposition and interpret an inverse column

The column-sum hypothesis guarantees $I - C$ invertible for a nonnegative consumption matrix. For instance if $q < 1$ is the largest column sum, $z = Cz$ would give $\|z\|_1 \leq q\|z\|_1$, forcing $z = 0$; the square-matrix theorem then gives invertibility.

(a) The given equations are $(I-C)x = d$ and $(I-C)\Delta x = \Delta d$. Adding them gives $(I-C)(x+\Delta x) = d + \Delta d$. Uniqueness makes $x + \Delta x$ the new production vector.

(b) If $\Delta d = e_1 = (1, 0, \ldots, 0)^T$, then $\Delta x = (I - C)^{-1}e_1$. Multiplication by $e_1$ selects column 1 of any matrix, so it selects exactly the claimed inverse column. The same interpretation applies to column $j$ for an increase in sector $j$'s final demand.

### Exercise 2.6.9: Solve a three-sector numerical production model

Multiply $(I - C)x = d$ by 10:

$$8x_1 - 2x_2 = 400, \qquad -3x_1 + 9x_2 - 3x_3 = 600, \qquad -x_1 + 8x_3 = 800.$$

The third equation gives $x_1 = 8x_3 - 800$. Substituting into the first gives $x_2 = 32x_3 - 3400$. Insert both expressions in the middle equation:

$$-3(8x_3 - 800) + 9(32x_3 - 3400) - 3x_3 = 600.$$

Thus $261x_3 = 28800$, giving $x_3 = 3200/29$, $x_1 = 2400/29$, and $x_2 = 3800/29$.

**Conclusion.** $x = (2400/29, 3800/29, 3200/29)^T$; direct substitution checks all three equations.

### Exercise 2.6.10: A demand increase affects every sector when every inverse entry is positive

An increase $h > 0$ in final demand for sector $j$ is $\Delta d = he_j$. The production change is $\Delta x = h(I - C)^{-1}e_j$, namely $h$ times inverse column $j$. Every entry of that column is strictly positive by hypothesis.

**Conclusion.** Production must rise in every sector, including sectors whose final demand did not change directly. Intermediate purchases transmit the increase throughout this model.

### Exercise 2.6.11: Verify the two GDP expressions algebraically

The production equation gives $d = (I-C)x$. The dual price equation gives $v = p-C^Tp$, so transposing yields $v^T = p^T - p^TC = p^T(I - C)$. Therefore

$$p^Td = p^T(I - C)x = v^Tx.$$

This uses associativity and the transpose identity $(C^Tp)^T = p^TC$; all three expressions are scalar values. It proves the equality without assuming an inverse for either equation.

### Exercise 2.6.12: An iterative inverse from finite geometric sums

By definition $D_m = I + C + \cdots + C^m$. Multiplying by $C$ shifts every power up by one, so

$$D_{m+1} = I + CD_m, \qquad D_0 = I.$$

Equivalently $D_{m+1} - D_m = C^{m+1}$. Thus repeatedly multiply the current matrix by $C$ and add $I$ to obtain the next approximation. To verify the limit, the finite telescoping identity is $(I - C)D_m = I - C^{m+1}$. The hypothesis $C^m \to 0$ implies $I - C$ invertible: if $z = Cz$, then $z = C^mz \to 0$, so $z = 0$. Consequently

$$D_m = (I - C)^{-1}(I - C^{m+1}) \longrightarrow (I - C)^{-1}.$$

The recurrence therefore approximates the requested inverse, not just an unexplained matrix sequence.

### Exercise 2.6.13: Compute production for the supplied seven-sector economy

**Enter the supplied consumption data exactly.**

To specify the data without ambiguous rounding, write

$$C = \frac{1}{10000}\begin{bmatrix} 1588 & 64 & 25 & 304 & 14 & 83 & 1594 \\ 57 & 2645 & 436 & 99 & 83 & 201 & 3413 \\ 264 & 1506 & 3557 & 139 & 142 & 70 & 236 \\ 3299 & 565 & 495 & 3636 & 204 & 483 & 649 \\ 89 & 81 & 333 & 295 & 3412 & 237 & 20 \\ 1190 & 901 & 996 & 1260 & 1722 & 2368 & 3369 \\ 63 & 126 & 196 & 98 & 64 & 132 & 12 \end{bmatrix}.$$

Use the demand vector $d = (74000, 56000, 10500, 25000, 17500, 196000, 5000)^T$ in the same sector order as the problem. Solve $Mx = d$ with $M = I - C$; this is a seven-equation linear system, not a demand for external data.

**Solve and check the residual before rounding.**

Exact rational elimination gives the following values, rounded to six decimal places for display (units: millions of dollars):

$$x \approx \begin{bmatrix} 99575.653398 \\ 97703.022863 \\ 51230.523166 \\ 131569.921929 \\ 49488.491372 \\ 329554.452570 \\ 13835.335715 \end{bmatrix}.$$

The exact rational solution satisfies $Mx - d = 0$ in every coordinate. The six-decimal display is only an approximation and need not have an identically zero numerical residual. To reproduce that exact calculation, define $C$ using the integer matrix above divided by 10000, and run:

```
from sympy import Matrix, eye
# C is the displayed integer matrix divided by 10000.
d = Matrix([74000,56000,10500,25000,17500,196000,5000])
M = eye(7) - C
x = M.LUsolve(d)      # exact rational elimination
assert M*x == d
print(x.evalf(12))
```

This solves the system without constructing an inverse. The displayed vector is the actual production plan, not merely an algorithm for finding it.

### Exercise 2.6.14: Use the same coefficient matrix for a new demand

Keep the consumption matrix from Exercise 2.6.13 and change only the demand to

$$d = (99640, 75548, 14444, 33501, 23527, 263985, 6526)^T.$$

Solving $(I - C)x = d$ by exact arithmetic, then displaying six decimals, gives (again in millions of dollars)

$$x \approx \begin{bmatrix} 134033.615328 \\ 131686.642968 \\ 69471.699079 \\ 176912.002023 \\ 66595.556706 \\ 443772.902077 \\ 18431.177683 \end{bmatrix}.$$

The same reproduction code works with this new $d$. Checking the exact solution gives $(I-C)x-d = 0$ coordinate by coordinate. A previously computed $LU$ factorization of $I - C$ can be reused; only its two triangular solves need be repeated.

### Exercise 2.6.15: Count the actual number of iteration steps

Start with the specified $x^{(0)} = d$ from Exercise 2.6.13 and iterate $x^{(k+1)} = d + Cx^{(k)}$ without rounding intermediate values. We interpret "to four significant figures" as agreement, in all seven coordinates,

with rounding the exact solution to four significant figures. The resulting target vector is

$$\begin{bmatrix} 9.958 \times 10^4 \\ 9.770 \times 10^4 \\ 5.123 \times 10^4 \\ 1.316 \times 10^5 \\ 4.949 \times 10^4 \\ 3.296 \times 10^5 \\ 1.384 \times 10^4 \end{bmatrix}.$$

Actual iteration gives these checks:

| $k$ | $\max_i \lvert x_i^{(k)} - x_i \rvert / \lvert x_i \rvert$ | all rounded entries agree? |
|---|---|---|
| 0 | $8.09987 \times 10^{-1}$ | no |
| 5 | $4.04043 \times 10^{-2}$ | no |
| 10 | $1.62676 \times 10^{-3}$ | no |
| 15 | $6.49063 \times 10^{-5}$ | no |
| 16 | $3.40686 \times 10^{-5}$ | yes |

.

At step 16, before rounding,

$$x^{(16)} \approx \begin{bmatrix} 99575.105594 \\ 97701.855867 \\ 51228.777814 \\ 131566.362702 \\ 49487.009311 \\ 329550.480997 \\ 13835.050858 \end{bmatrix}.$$

No preceding iterate has all seven required rounded values. At step 15, for instance, the first coordinate is 99574.609463, which rounds to $9.957 \times 10^4$, not $9.958 \times 10^4$. Here is a reproducible counting procedure using the exact $C, d$:

```
x_exact = (eye(7)-C).LUsolve(d)
four = lambda v: [format(float(z), '.4g') for z in v]
target = four(x_exact)
xk = d
k = 0
while four(xk) != target:
    xk = d + C*xk
    k += 1
assert k == 16
```

The rational arithmetic prevents iteration-roundoff accumulation; the final formatting is used only for the specified comparison.

**Conclusion.** With $x^{(0)} = d$ counted as step zero, 16 updates are needed. The usual relative-error check $\max_i \lvert x_i^{(k)} - x_i \rvert / \lvert x_i \rvert < 5 \times 10^{-5}$ also first holds at step 16 for these data.